

Sreelekshmi Sreekumar

sreelekshmi.sreek@gmail.com | U.S. Citizen | sreelekshmisreekumar.com

Education

University of Central Florida (UCF)

Bachelor of Science in Mechanical Engineering, Bachelor of Science in Computer Science

August 2020 – August 2025

Overall GPA: 3.62

Work Experience

General Dynamics Mission Systems

Scottsdale, Arizona, USA

Manufacturing Engineer II

June 2026 - Present

- Execute Build Readiness Reviews (BRRs) to verify manufacturing readiness, process compliance, and production flow for space communication hardware
- Provide floor support to technicians and operations by interpreting and applying IPC IPC-A-610 and J-STD-001 requirements to resolve manufacturing issues and ensure hardware workmanship compliance
- Troubleshoot manufacturing process issues by developing and executing test plans, performing root cause analysis, and implementing corrective actions to improve yield, throughput, and product reliability

Exolith Lab/Florida Space Institute

Orlando, Florida, USA

Lead Research/Mechanical Engineer

February 2022 – July 2025

- Led DFM/DFA-driven design, fabrication, and integration of a solar-powered regolith sintering system, delivering a full-scale $5 \times 5 \times 7$ ft prototype to demonstrate structural feasibility, automation concepts, and repeatable sintering capability
- Designed and integrated structural frames, motion systems, and actuators using SolidWorks, additive manufacturing, and machining to enable controlled positioning, repeatable test motion, and subsystem interoperability during operations
- Performed 12 MTS compression and three-point bend tests on sintered samples, identifying excessive porosity and brittle failure behavior that constrained load-bearing performance
- Developed a low-cost regolith pre-compaction process, experimentally demonstrating reduced porosity and improved material consistency after sintering
- Restored test readiness by diagnosing 15 mechanical, electrical, and software failures using root cause analysis during operations and implementing corrective design changes to reduce unplanned downtime
- Coordinated onboarding and technical training for 5 interns, including the new leader of the project, for 3 months
- Authored system operations and test documentation, as well as an AIAA research paper

NASA Ames Research Center

Mountain View, California, USA

Mechanical Test Engineering Intern

August 2023 – December 2023

- Supported vibration, functional, and environmental test planning through fixture concept development on PTC Creo
- Designed a welded aluminum and 3D-printed mass simulator using hand calculations and free-body diagrams to meet required mass and center-of-gravity targets, balancing structural integrity, distortion control, cost, and fabrication time
- Designed a vibration test adapter plate with CG-balanced geometry, threaded inserts, and mounting patterns compatible with facility vibration tables to reduce test-induced vibration artifacts and center-of-gravity shifts

Vasu Lab (Combustion, Propulsion, Laser Diagnostics and Absorption Sensors, and Fuels)

Orlando, Florida, USA

Research Assistant

May 2023 – February 2025

- Designed Schlieren imaging layouts under injector, optical access, and sensor interference constraints to enable non-intrusive visualization of high-speed flow structures; published data in a paper for AIAA
- Processed data using Python and MATLAB to extract density gradients for qualitative assessments of flow behavior

Projects

Talon Simulations F-35 Flight Simulator

January 2025 – August 2025

- Drove the design, documentation, integration, and testing of an F-35 throttle control system for a pilot training simulator, creating a custom SolidWorks throttle mechanism, electrical architecture, and Python-based control software
- Authored 6 technical documents defining 21 high-level system requirements and validation test plans, and conducted a system-level FMEA identifying 20 potential failure modes to mitigate risk and improve overall reliability

Students for the Exploration and Development of Space at UCF Projects

August 2020 – August 2021

- Designed a six-legged planetary rover chassis in SolidWorks for NASA Big Ideas and performed FMEA on a liquid bi-propellant rocket propulsion system to identify failure modes and define risk mitigation strategies

Skills

Software: C, C++, Java, JavaScript, Python, MATLAB, Ubuntu, PTC Creo, SolidWorks, LabVIEW, OpenRocket, GitHub, Jira

Technical: GD&T, FEA, machining, additive manufacturing, woodworking, encoders, microcontroller, DAQ, forklift